

Rahul Jana

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Machine Learning Engineer with 3+ years building and deploying end-to-end ML and computer vision systems on AWS. Designed scalable MLOps pipelines with 98% uptime, optimized computer vision models(eg. YOLO, UNet) for production, and contributed to open-source research with published work. Finalist in a \$1.4M global AI innovation challenge.

Experience

TIF- AWADH, Indian Institute of Technology - Ropar

Chandigarh

Software Engineer

April 2023 - Present

- Designed and operated AWS-based ML deployment pipelines with **98% uptime** using Docker, SageMaker, and CI/CD workflows.
- Designed an accelerometer-based cattle activity detection model achieving **93% accuracy** (*published on arXiv*).
- Built an AI-driven pest surveillance system using the KCC (Kisan Call Center) queries, to enable early pest outbreak detection and decision support for the government and farmers. Achieved an average **lead time of 2 weeks** for outbreak detection.
- Developed a computer vision and attention-based solution for predicting cattle body attributes (**RMSE: 8 cm $\pm$ 2**) and body weight (**RMSE: 10 kg $\pm$ 3**). helping dairy researchers and farmers track their cattle's growth.
- Built a U-Net and VGG16 encoder-based segmentation and style-transfer system for automated interior design.

Springworks

Remote

Machine Learning Intern

September 2021 - July 2022

- Engineered machine learning models enabling document quality detection across government ID formats (Aadhar, PAN, Passport), achieving **95%** accuracy with a false positive rate under 5%.
- Improved runtime performance of a crypto trading algorithm by **~20%** using live trading scenario.

Open source Contributions, Projects & Hackathons

Deep Forest Sciences

- Finalist in **Standard Industries Chemical Innovation Challenge for Advancement of AI-Assisted Molecular Synthesis**, for the prize pool of \$1.4 Million.
- Contributed in backend development for **DeepRetro, a Retrosynthetic Pathway Discovery using Iterative LLM Reasoning [GitHub]**(*published on arXiv*).

Voice-enabled RAG chatbot

- Built a production-grade RAG pipeline using FAISS with OpenAI embeddings and GPT-3.5-turbo, implementing fully async query execution to eliminate event-loop blocking under concurrent requests.
- Implemented Hinglish-aware transcription normalization to enable accurate semantic retrieval across Hindi, English, and Hinglish documents with structured endpoint logging, and graceful database degradation, maintaining system availability even on DB failure.

Skills

ML and GenAI: • Python • PyTorch • LangChain • Prompt Engineering • RAG • MCP • HuggingFace
Cloud and MLOps: • AWS • Docker • PostgreSQL • MongoDB • MLflow • FastAPI • GitHub Actions

Education

Chandigarh University

Punjab, India

Bachelor of Engineering with Specialization in AI and ML(CGPA: 7.35 / 10.0)

- Developed a Voice-Controlled Switch Box for **Socially Relevant Projects** and **Abdul Kalam Innovation Conclave**.